COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE-010 - Data Structures and Algorithms	**SECTION:**	CPE21S2
DATE PERFORMED:	9/18/26	**DATE SUBMITTED:**	9/18/26
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Assignment 9.1 **Binary Search Tree**

1. PROCEDURE OUTPUT

1. What is a binary search tree?
   - A BST is a type of binary tree where every node's left child and descendants have lower values than the node itself, while its right child and descendants have higher values. This organized structure enables fast searching, inserting, and deleting values that can be done in O(log n) time for a balanced BST without shifting elements in memory.
2. Where can binary search trees be used?
   - BSTs are widely used in applications requiring fast searching and dynamic data management, such as database indexing systems, file systems, and autocomplete features, and expression evaluation in compilers. They're also employed in implementing sorted data structures, priority queues, and memory allocation algorithms.
3. What is a tree traversal?
   - Tree traversal is the process of systematically visiting and processing each node in a tree in a specific order. There are several traversal methods such as pre-order (visiting a node before its children), in-order (visiting between left and right children), and post-order (visiting after both children), each suited for different applications, so the choice of traversal method depends on the task, as different orders produce different results.
4. Differentiate a post-order and pre-order traversal give examples.
   - A post-order traversal works from the bottom-up. It goes Left→Right→Root from the smallest subtree to the biggest. This is usually used for evaluation of expressions.
   - A pre-order traversal works from the root to the leftmost, then right. It goes Root→Left→Right from the biggest to the lowest branch, then back up. It can be used when copying a tree.

5. What is a parse tree and its use?
   - A parse tree is a hierarchical representation of a string/expression written in the code or inputted by the user. It is used in compilers and interpreters to validate syntax. In simpler terms, it divides the string into branches and puts them in order of operation so that calculations can be calculated as it should be, the best example being order of mathematical operations (PEMDAS). It usually uses Post-order traversal to evaluate expressions and In-order traversal to turn it into readable format.
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2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Copy and paste the questions from the lab and include your answers after.

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3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

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